Andreas Demosthenous

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Summary

CS undergraduate at the University of Cyprus (2023–2027), an information-security (GRC) intern at Eurobank Cyprus, and a Software Engineer intern at KIOS. Work spans geospatial Python, full-stack web, embedded firmware, low-level Linux, self-hosted infrastructure, security and compliance, and on-device LLM inference.

Experience

Information Security Intern, Eurobank Cyprus – Nicosia, Cyprus — June 2026 – present

- Information security internship on the governance, risk, and compliance (GRC) team.

Software Engineer Intern, KIOS Research and Innovation Center of Excellence – Nicosia, Cyprus — Jan 2025 – present

- Develop Python / QGIS plugins for geospatial water-network modelling; design data-ingestion pipelines turning tabular utility datasets into topologically correct GIS layers.
- Set up CI/CD for in-house plugin distribution using GitHub Actions and Git LFS.

Education

University of Cyprus, BSc in Computer Science – Nicosia, Cyprus — Sept 2023 – June 2027

- Coursework: Operating Systems, Computer Architecture, Distributed Systems, Algorithms, Cryptography, Software Engineering.

Projects

Hisense A/C de-cloud — custom Matter firmware (open source)

- Reverse-engineered a Hisense AEH-W41H1 A/C Wi-Fi module (Realtek RTL8710C / AmebaZ2): sniffed and decoded its internal RS-485 protocol, then wrote custom Matter firmware in C/C++ that replaces the vendor cloud with fully local Home Assistant control (zero cloud).
- Shipped the whole pipeline: CH341 SPI flashing plus Wi-Fi OTA, host-only QA (codec and protocol round-trip tests), and GitHub Actions that build, sign, and publish firmware releases; includes an ESP32 (esp-matter) replacement path. github.com/AndrewDemsDS/hisense-w41h1

Self-hosted homelab (Docker, ~25 services)

- Run a 25-service Docker Compose stack (Nextcloud, Vaultwarden, Paperless-ngx, Jellyfin, Gitea, Home Assistant) behind Traefik with Let's Encrypt and a Cloudflare Tunnel; migrated it from a Raspberry Pi 5 to an x86 QuickSync NAS.
- Ran a security audit and hardened the stack (socket-proxied Watchtower, no-new-privileges, reverse-proxy auth, secret rotation, key-only SSH, restic encrypted backups), then segmented a two-site UniFi network with VLANs, a zone-based firewall, and a WireGuard site-to-site mesh.

baseplate (open source)

- Opinionated single-host self-hosting template with hardened defaults (socket-proxied Watchtower, no-new-privileges, security headers), Traefik, and a Cloudflare Tunnel. github.com/AndrewDemsDS/baseplate

Smart-home automation and custom security system

- Multi-node ESP32 / Raspberry Pi system integrating sensors, actuators, and a custom security layer; full-stack web UI for monitoring and control.

dotfiles (open source)

- Hyprland + Quickshell desktop (fork of illogical-impulse) with custom QML widgets: Home Assistant control, a VPN/network pill, an RSS reader, and a homelab glance panel. github.com/AndrewDemsDS/dotfiles

Skills

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, Bash

Web & data: Full-stack development, PostgreSQL, REST APIs

Geospatial: PyQGIS, QGIS plugin development, EPANET tooling, pandas/openpyxl

Systems: Linux (Gentoo, NixOS, Arch), Android device porting, systemd, kernel building

AI / LLMs: Self-hosted inference (Ollama, llama.cpp), MCP server development

Infrastructure: Docker / Compose, Traefik, Cloudflare Tunnels, restic, Nix

Networking: VLAN segmentation, UniFi, zone-based firewalls, WireGuard, site-to-site VPN

Hardware: ESP32, Raspberry Pi, Arduino

Spoken: Greek (native), English (B2+, English for Computer Science certification)